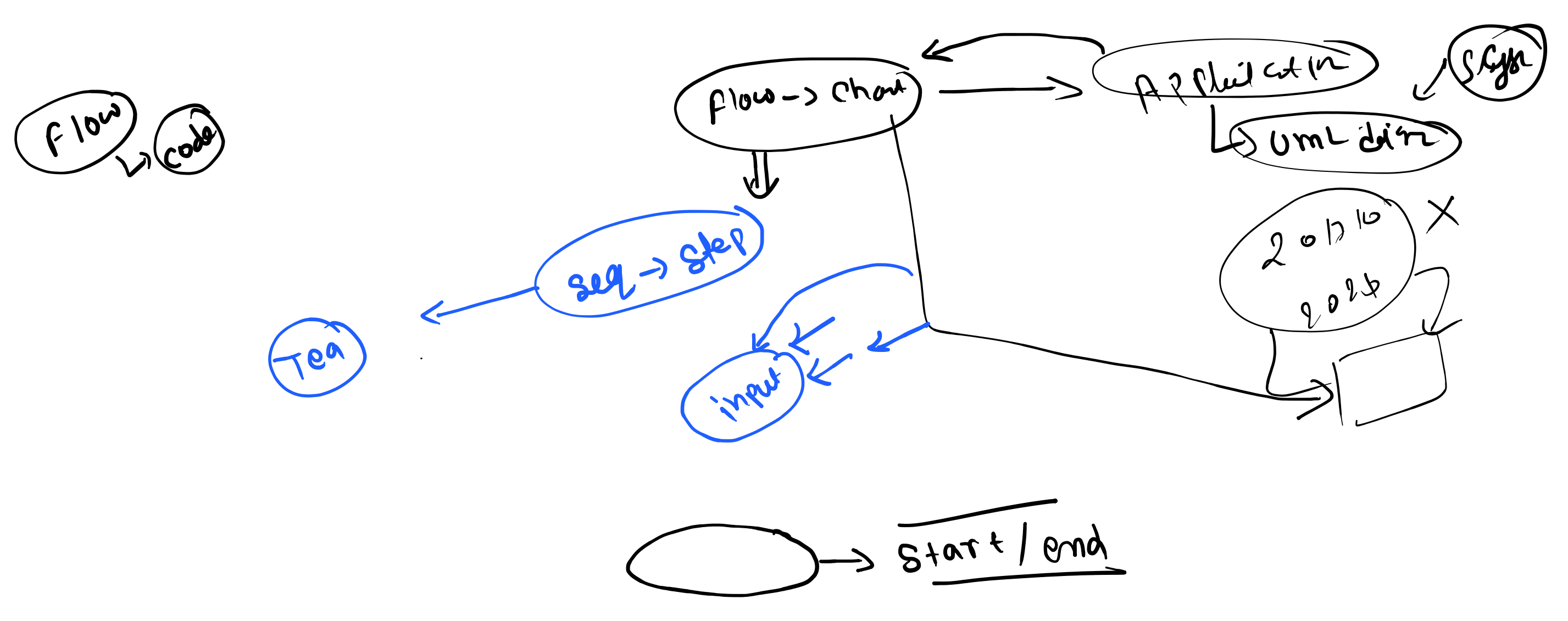
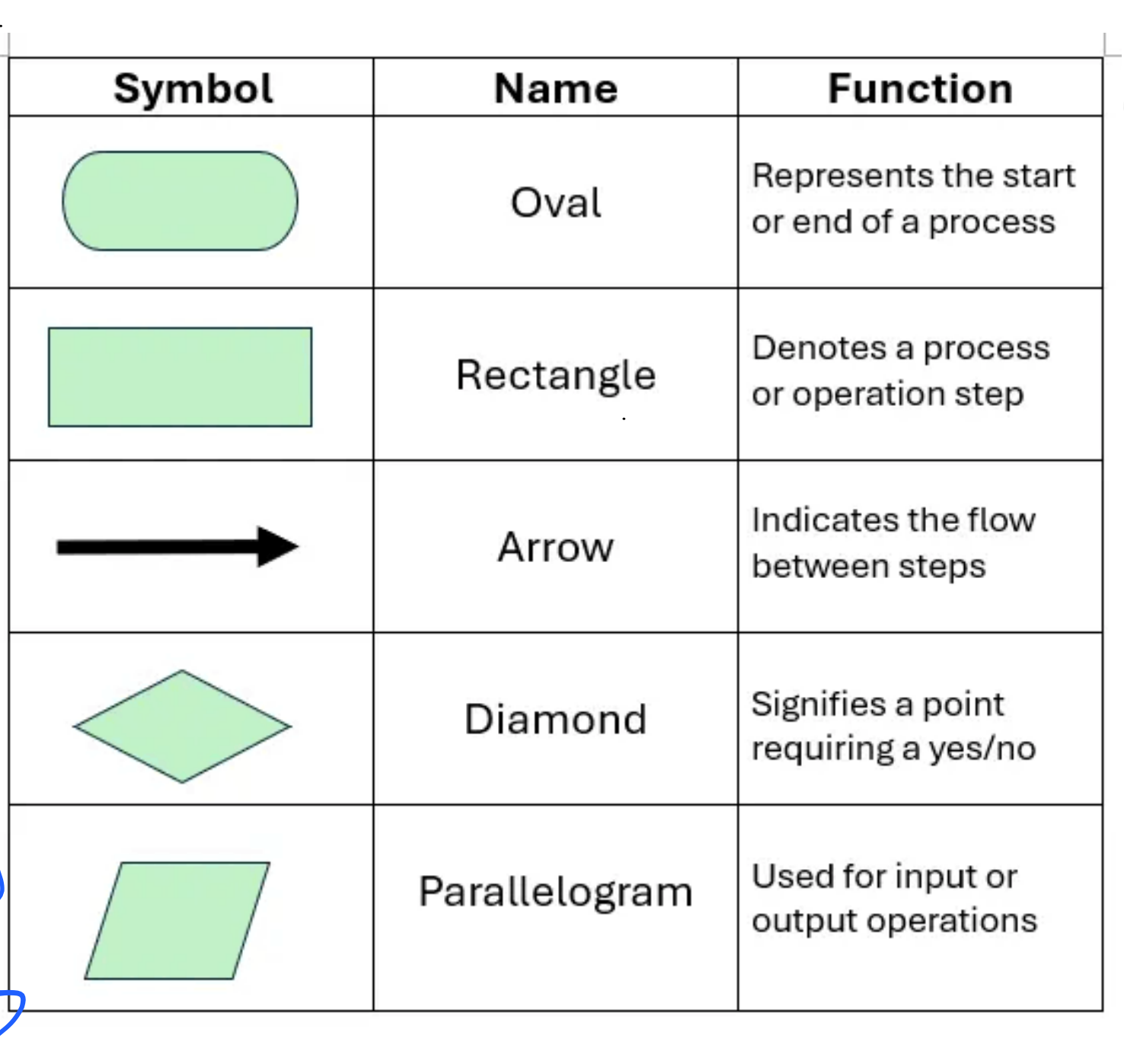
Flow-chart

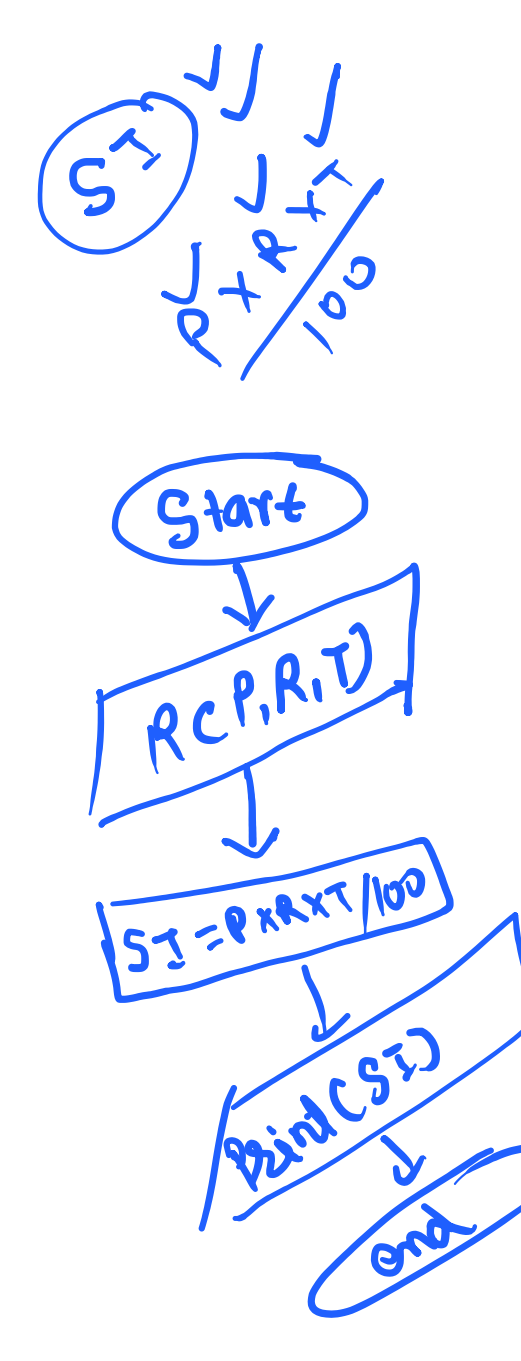
17/05/26 12:39 PM


Flow → code
Flow → chart
seq → step
Tea
input
Application
Sign
UML diagram
2016
2026
Start / end

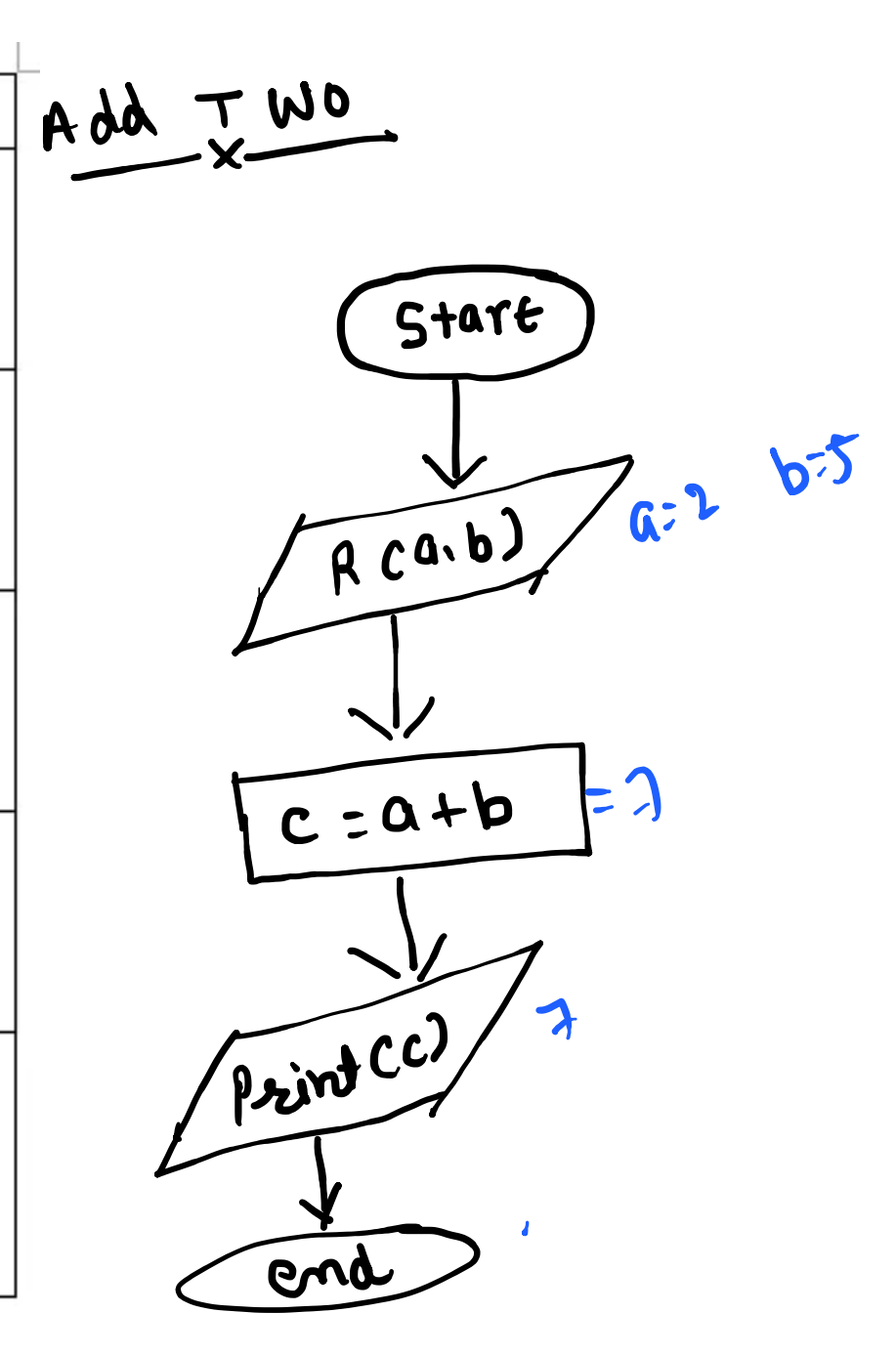


Symbol	Name	Function
🟩	Oval	Represents the start or end of a process
🟩	Rectangle	Denotes a process or operation step
→	Arrow	Indicates the flow between steps
◆	Diamond	Signifies a point requiring a yes/no
▱	Parallelogram	Used for input or output operations

SI ✓ P × R × T / 100


Start
R(P,R,T)
SI = P×R×T/100
Print(SI)
end

Add two


Start
R(a,b)
a=2 b=5
c = a+b
=7
Print(c)
7
end

①

odd/even

% ←

Rem = 0

71/8 = 8

5/8 = 0

11 % 8 = 3

71 % 8 = 7

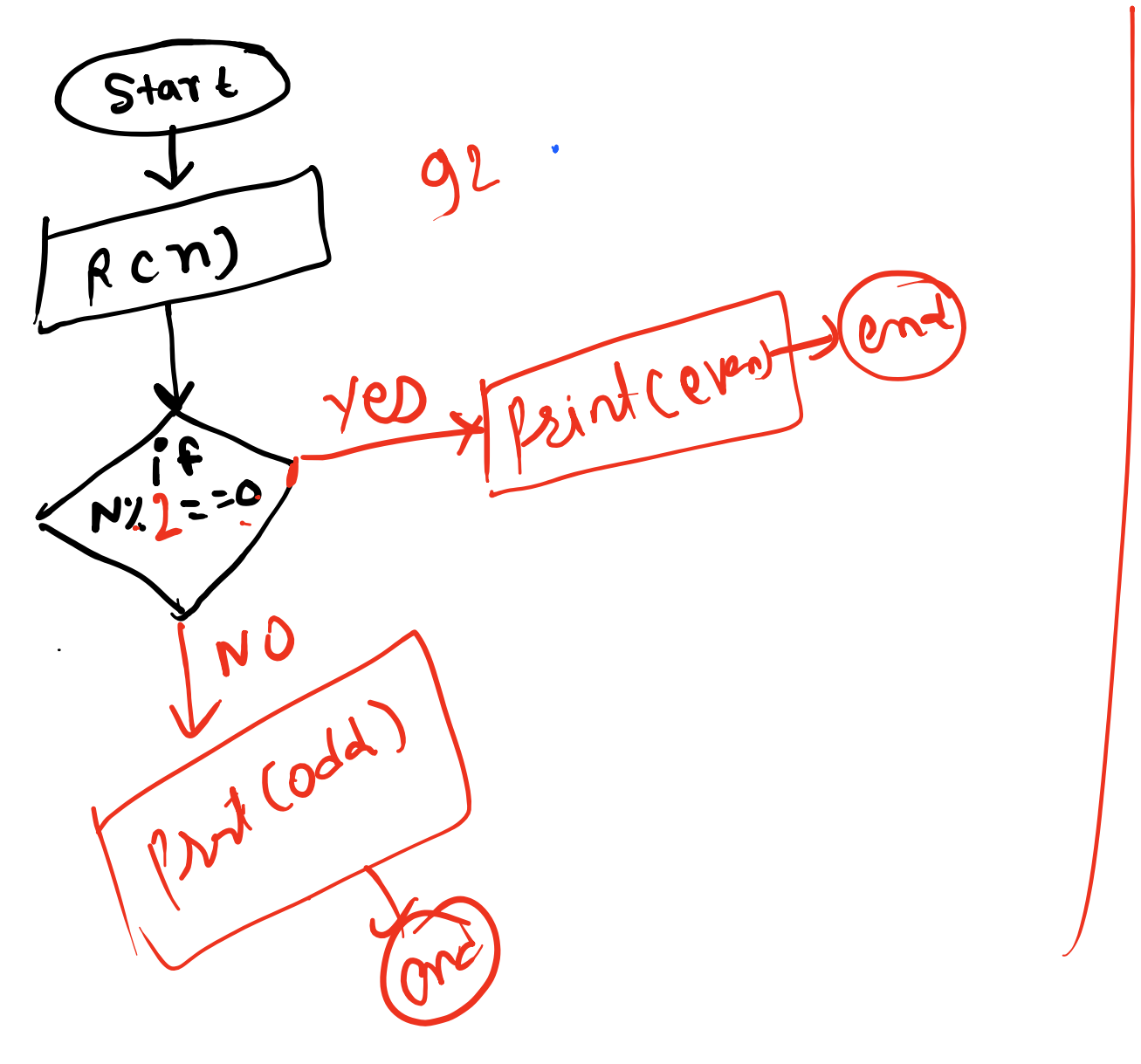
5 % 8 = 5 ✓

8)71 (8
64
7

-7 % 5 = -2 → +ve -2+5 = ③

-3 % 5 = -3 → +ve -3+5 = ②

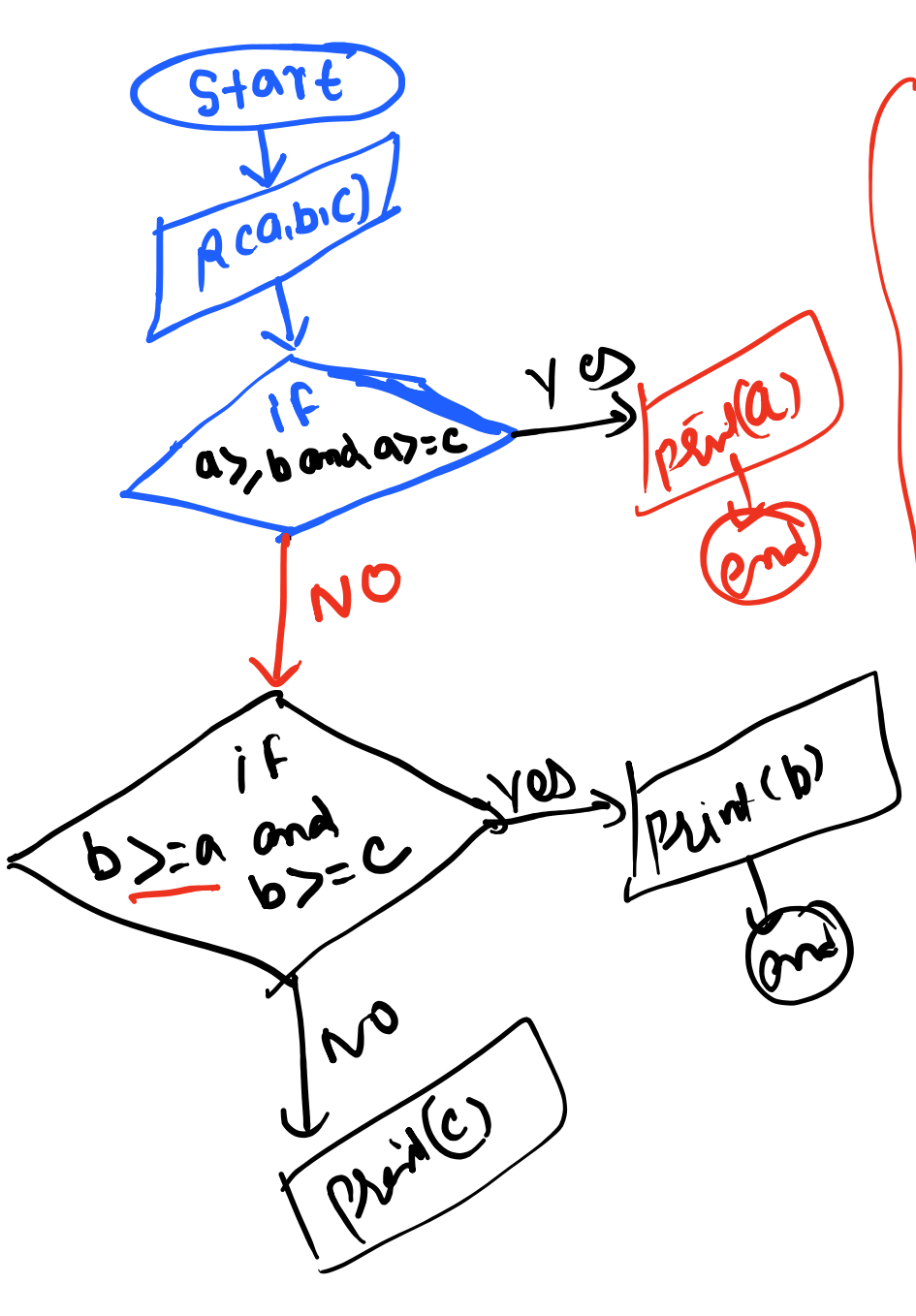
⑪

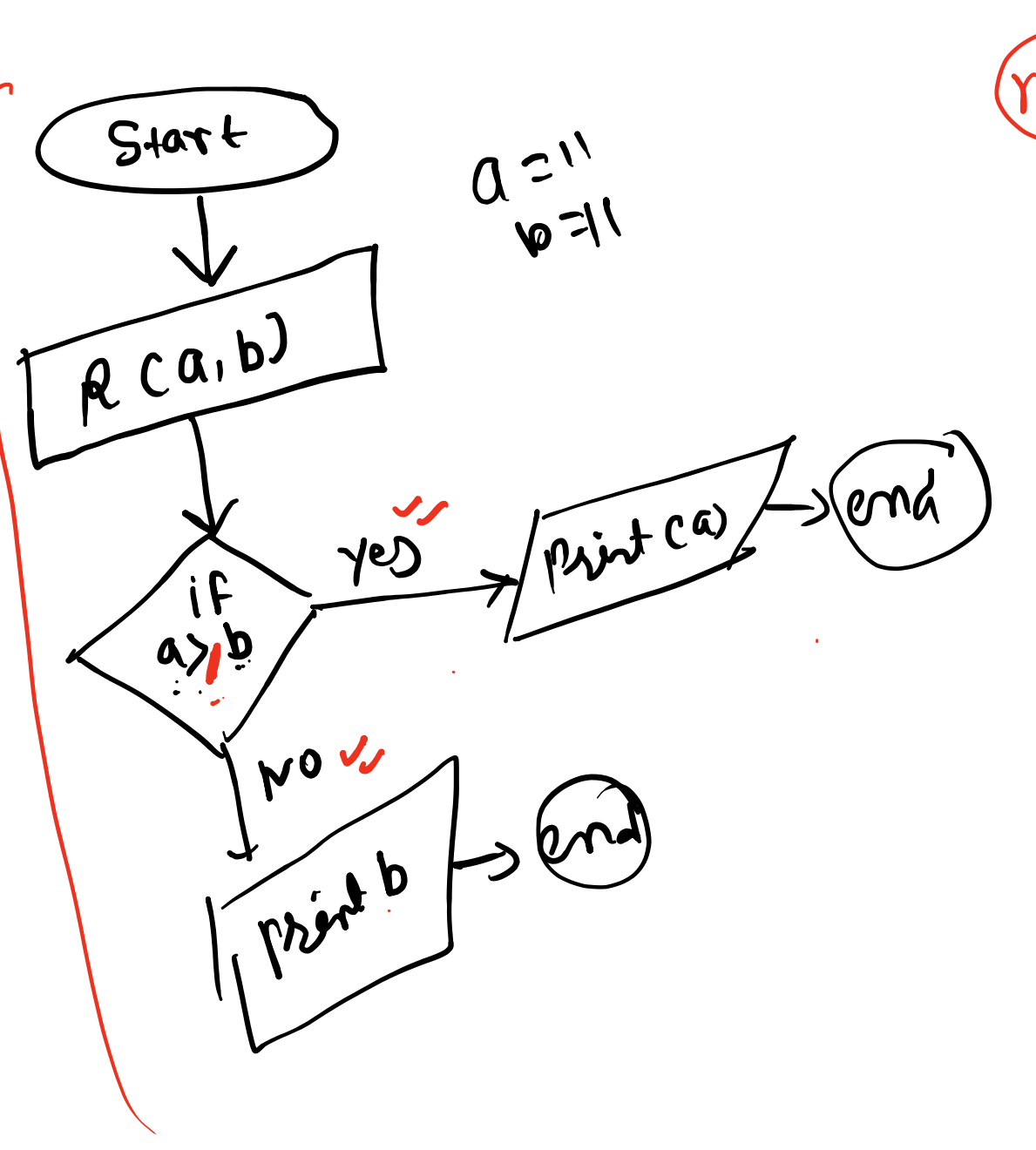

Start
R(n)
92
if N%2==0
yes
Print(even)
end
NO
Print(odd)
end

x = 5

x == 5

(a,b) → larger


Start
R(a,b,c)
if a>b and a>=c
yes
Print(a)
end
NO
if b>=a and b>=c
yes
Print(b)
end
NO
Print(c)


Start
R(a,b)
a=11
b=11
if a>b
yes
Print(a)
end
NO
Print b
end

n > 50

a,b,c

m = ✓

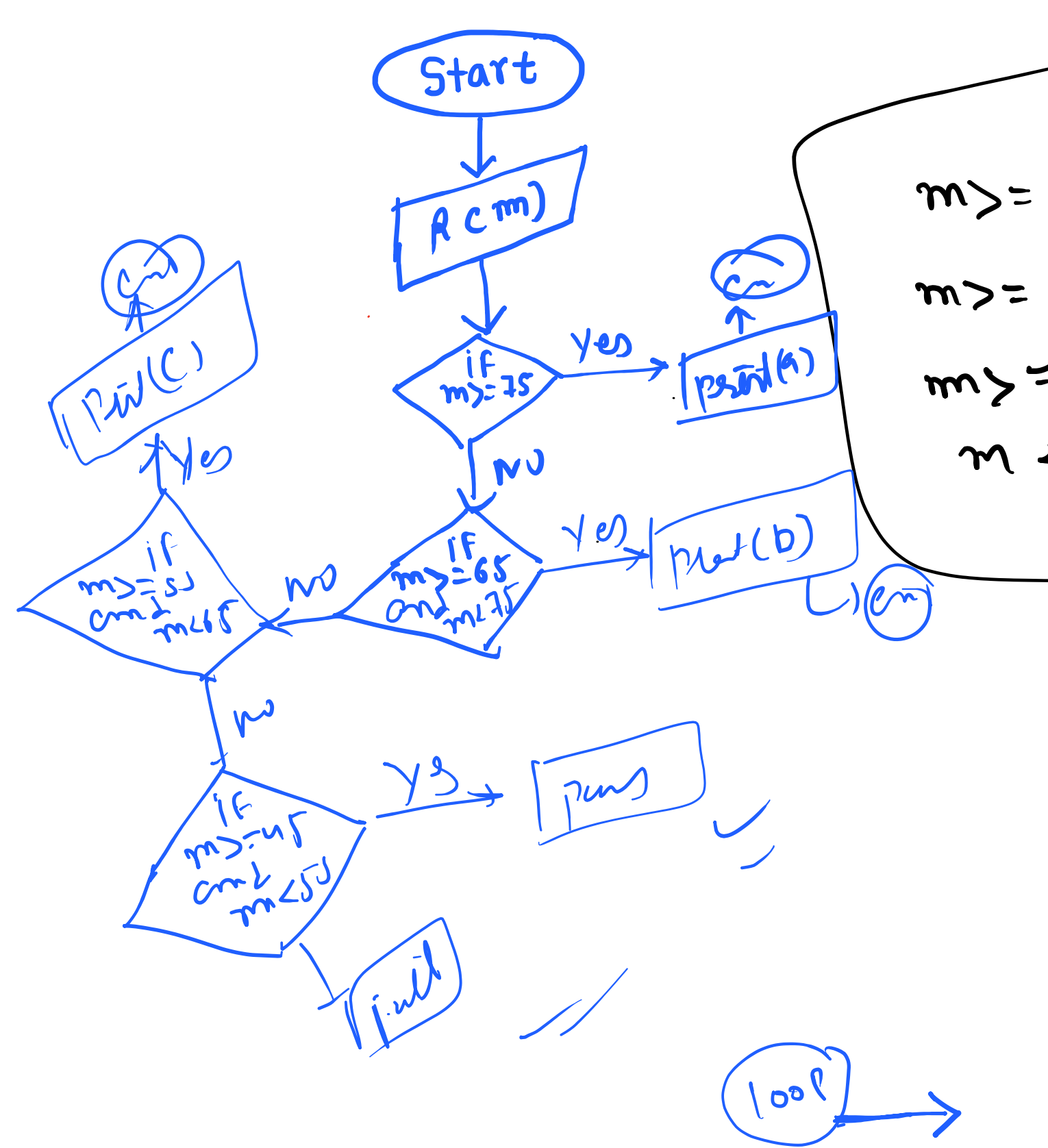
m >= 75 A

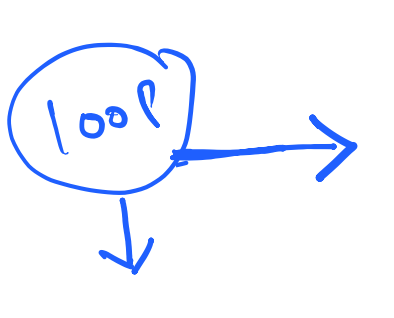
m >= 65 and m < 75 B

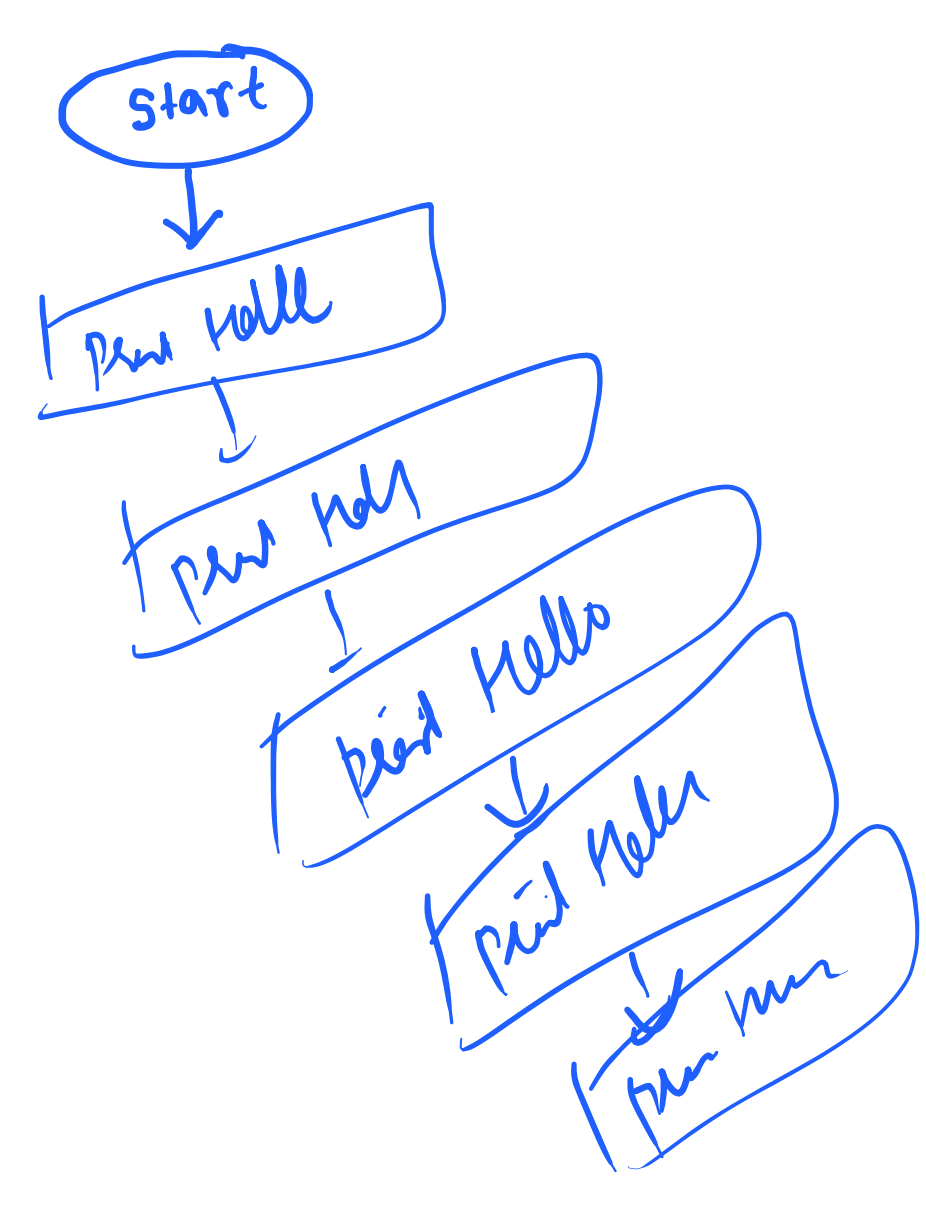
m >= 55 and m < 65 C

m >= 45 and m < 55 Pass

m < 45 Fail


Start
R(m)
if m>=75
Yes
Print(A)
end
NO
if m>=65 and m<75
Yes
Print(b)
end
NO
if m>=55 and m<65
Yes
Print(C)
end
NO
if m>=45 and m<55
Yes
pass
fail



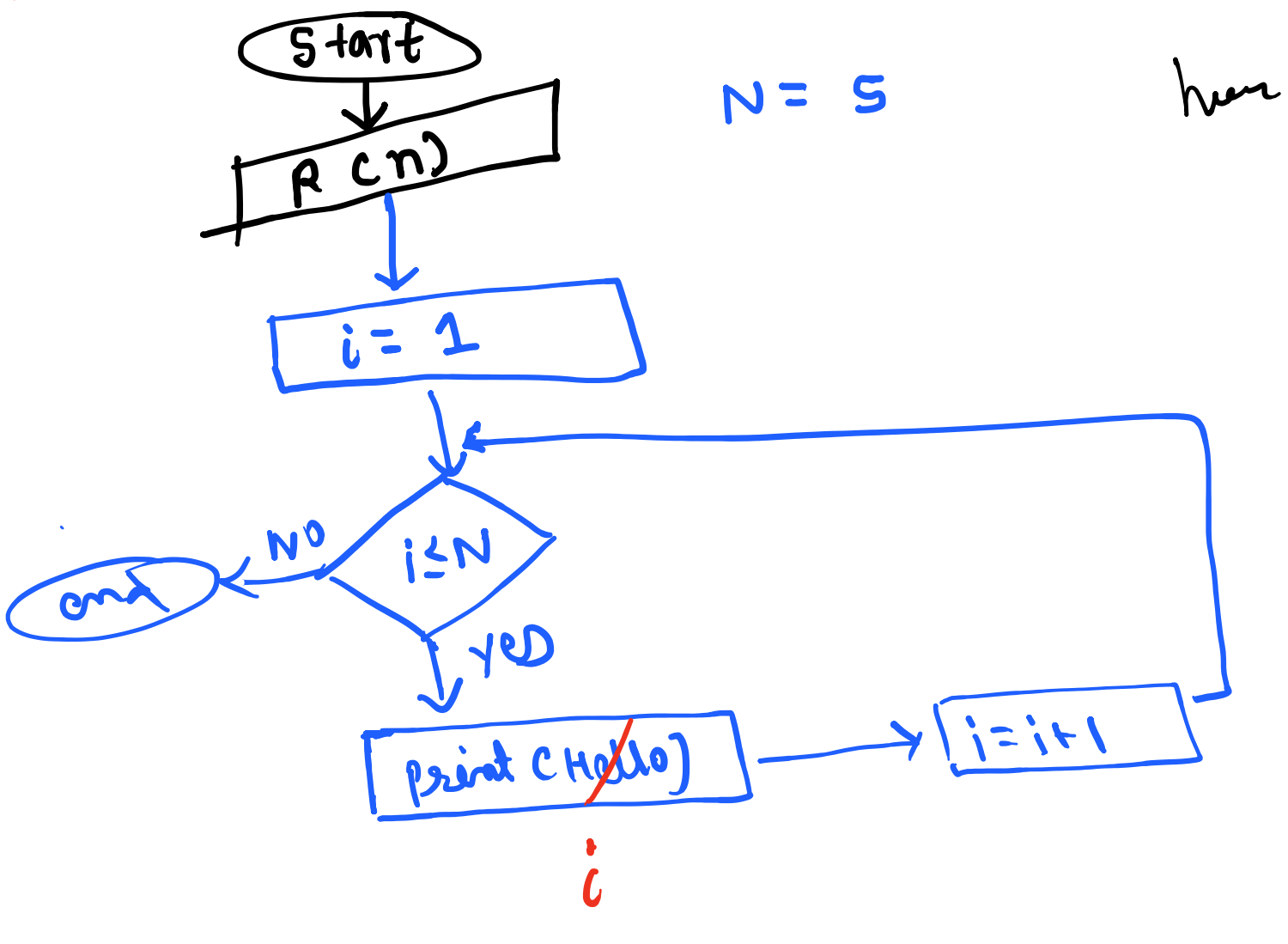

Start
Print Hello
Print Hello
Print Hello
Print Hello
Print Hello

Hello
Hello
Hello
Hello
Hello

n×(n+1)/2 ✗

n=5

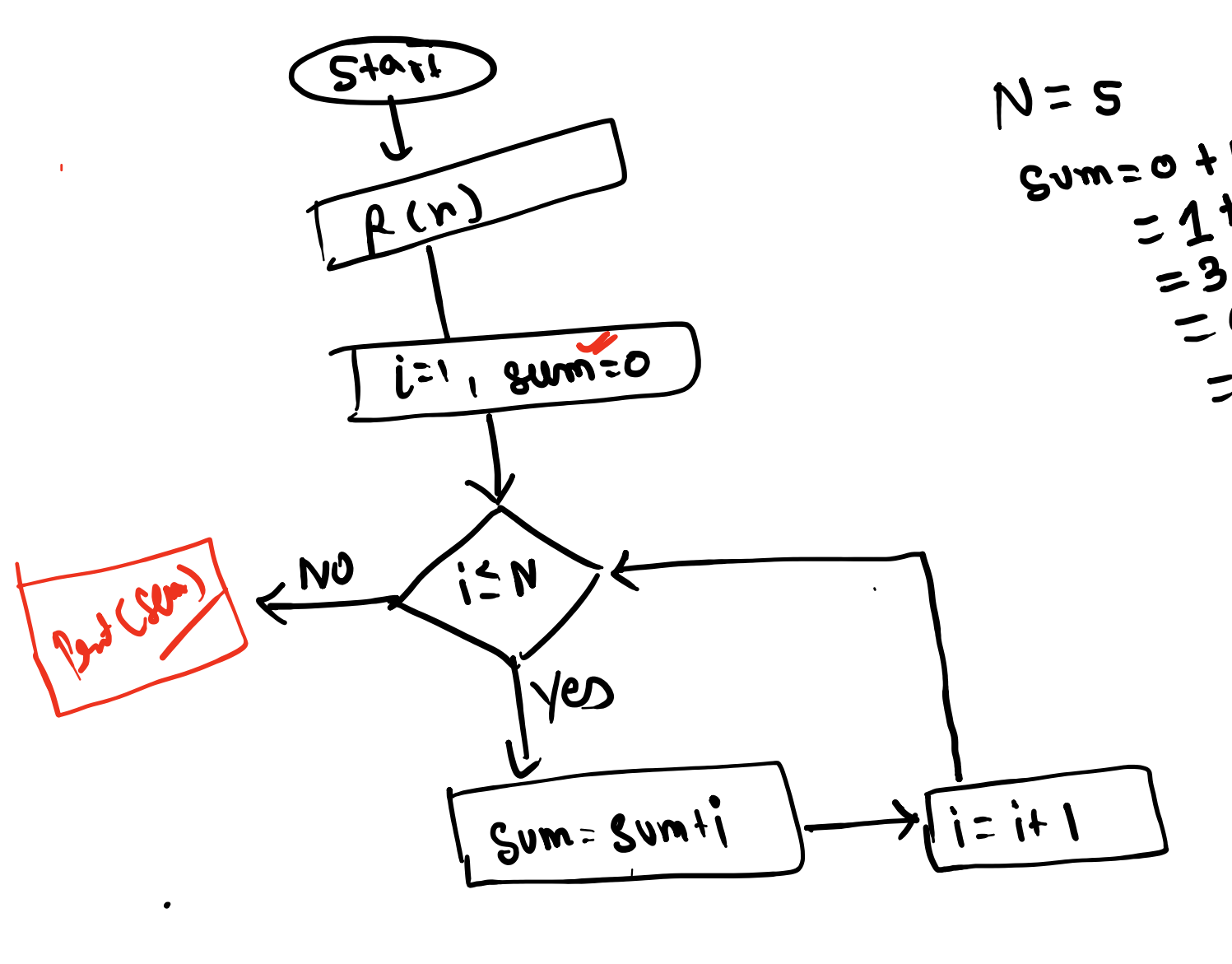
1+2+3+4+5 = 15


Start
R(n)
i = 1
i≤N
NO
end
yes
Print(Hello)
i
i = i+1

N = 5

how loop ✓

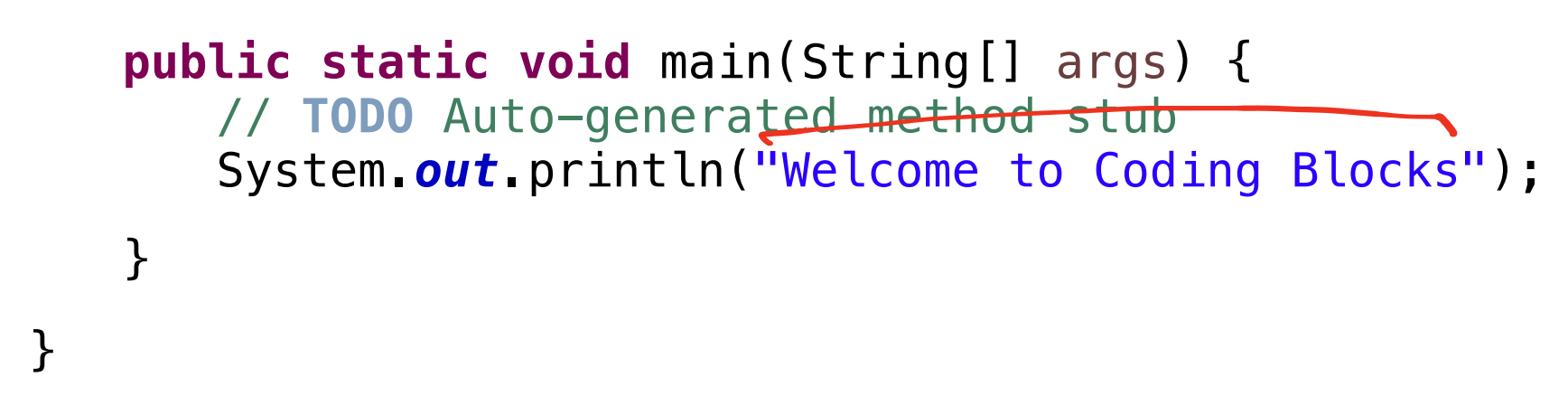
i=1 | 1≤5
i=2 | 2<=5
i=3 | 3<=5
i=4 | 4<=5
i=5 | 5<=5
i=6 | 6≤5


Start
R(n)
i=1, sum=0
i≤N
NO
Print(sum)
yes
sum = sum+i
i = i+1

N = 5

sum = 0+1
= 1+2
= 3+3
= 6+4
= 10+5
= 15

i = 1 | 1≤5
i = 2 | 2≤5
i = 3 | 3≤5
i = 4 | 4≤5
i = 5 | 5≤5
i = 6 | 6≤5

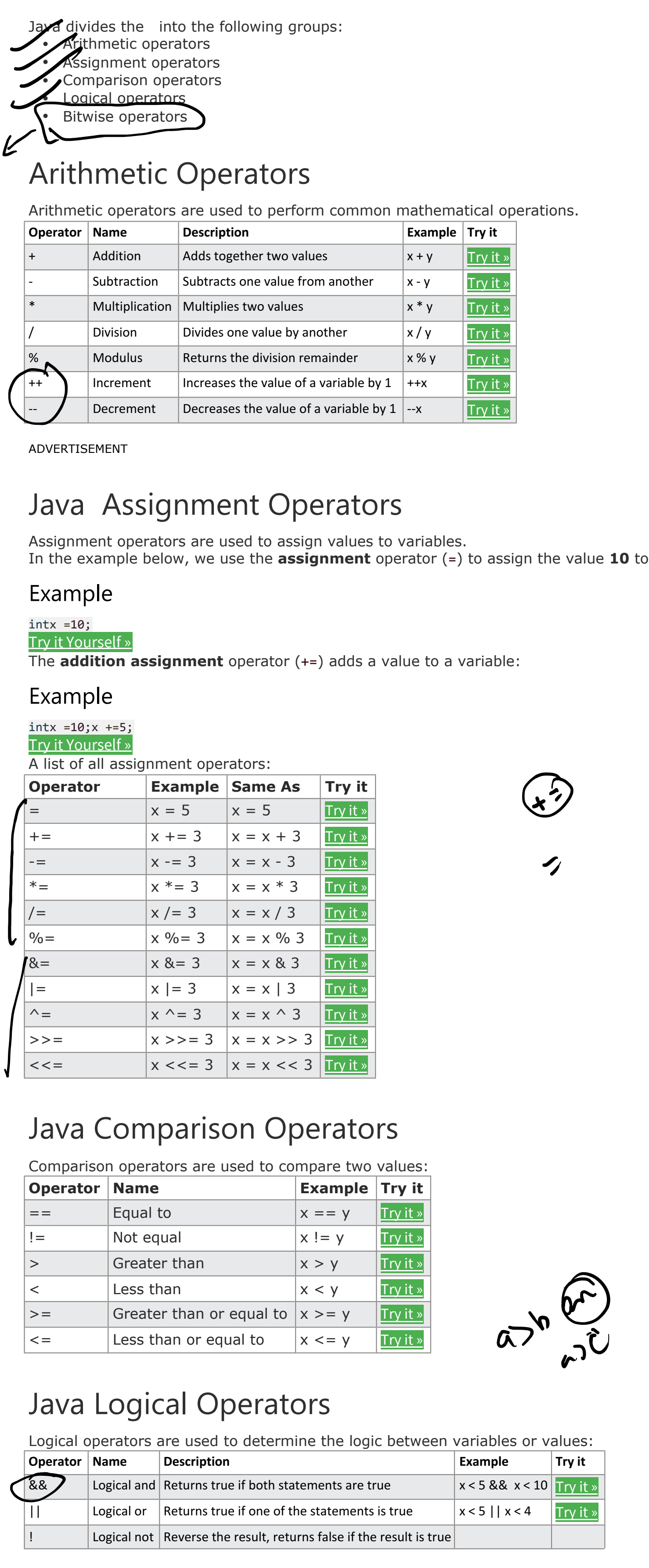


```
public class Test {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        System.out.println("Welcome to Coding Blocks");

    }

}
```



Java divides the into the following groups:

- Arithmetic operators
- Assignment operators
- Comparison operators
- Logical operators
- Bitwise operators

Arithmetic Operators

Arithmetic operators are used to perform common mathematical operations.

Operator	Name	Description	Example	Try it
+	Addition	Adds together two values	x + y	Try it »
-	Subtraction	Subtracts one value from another	x - y	Try it »
*	Multiplication	Multiplies two values	x * y	Try it »
/	Division	Divides one value by another	x / y	Try it »
%	Modulus	Returns the division remainder	x % y	Try it »
++	Increment	Increases the value of a variable by 1	++x	Try it »
--	Decrement	Decreases the value of a variable by 1	--x	Try it »

ADVERTISEMENT

Java Assignment Operators

Assignment operators are used to assign values to variables.
In the example below, we use the **assignment** operator (=) to assign the value **10** to a variable called **x**:

Example

`intx =10;`
Try it Yourself »
The **addition assignment** operator (+=) adds a value to a variable:

Example

`intx =10;x +=5;`
Try it Yourself »
A list of all assignment operators:

Operator	Example	Same As	Try it
=	x = 5	x = 5	Try it »
+=	x += 3	x = x + 3	Try it »
-=	x -= 3	x = x - 3	Try it »
*=	x *= 3	x = x * 3	Try it »
/=	x /= 3	x = x / 3	Try it »
%=	x %= 3	x = x % 3	Try it »
&=	x &= 3	x = x & 3	Try it »
|=	x |= 3	x = x | 3	Try it »
^=	x ^= 3	x = x ^ 3	Try it »
>>=	x >>= 3	x = x >> 3	Try it »
<<=	x <<= 3	x = x << 3	Try it »

Java Comparison Operators

Comparison operators are used to compare two values:

Operator	Name	Example	Try it
==	Equal to	x == y	Try it »
!=	Not equal	x != y	Try it »
>	Greater than	x > y	Try it »
<	Less than	x < y	Try it »
>=	Greater than or equal to	x >= y	Try it »
<=	Less than or equal to	x <= y	Try it »

Java Logical Operators

Logical operators are used to determine the logic between variables or values:

Operator	Name	Description	Example	Try it
&&	Logical and	Returns true if both statements are true	x < 5 && x < 10	Try it »
||	Logical or	Returns true if one of the statements is true	x < 5 || x < 4	Try it »
!	Logical not	Reverse the result, returns false if the result is true		

+
-
*
/
%

++ → (+1)

-- → (-1)

x = 5

x += 5 ⇒ x = x+5

x = x*5

x = 5

+=
-=
*=
/=
%=

x =+ 5

x =- 5

x += 5

x -= 5

x =* 5 ✗

(+=)

x = y

True x == y

false

x >= y

x > y

x <= y

x < y

a > b

a >= c

a	b	a||b
T	T	T
T	F	T
F	T	T
F	F	F

a	b	a&&b
T	T	T
T	F	F
F	T	F
F	F	F

x = 8 9 8 9 8 7

```
int a = x++ - --x + x++ - --x + --x;
```

$\cancel{8} - \cancel{8} + \cancel{8} - \cancel{8} + 7$

x = 8 9 8 9 8 9

```
int a = ++x + --x + x++ - --x + ++x + 2;
```

$= \underbrace{9 + 8 + 8 - 8 + 9}\ + 2$

$= \boxed{28}$